Abhinav Kumar Jha

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Education

Indian Institute of Information Technology (IIIT) Manipur — Manipur, India
Bachelor of Technology in Computer Science & Engineering — 2023 – 2027

- CGPA: 8.89 / 10.0
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Machine Learning

Experience

Dairywala — Remote
Tech Intern (Full-Stack) — Jun 2026 – Jul 2026

- Engineered full-stack modules for a dairy supply chain ecosystem, streamlining procurement, inventory tracking, operations, and analytics.
- Optimized RESTful APIs and database queries across live business workflows, identifying and resolving production bottlenecks.

IITianCraft — Remote
Backend Developer Intern — Oct 2025 – Dec 2025

- Spearheaded backend architecture for a Tally Automation Tool using Node.js and Express.js, building fault-tolerant XML/HTTP bridges between web applications and local Tally Prime servers.
- Automated real-time ledger and voucher synchronization, eliminating manual data entry errors and cutting accounting processing time by 40%.

Projects

VoiceCode – Agentic AI Coding Assistant | TypeScript, Node.js, Gemini API, Ollama, Jest — Oct 2026

- Built a voice-driven autonomous VS Code extension executing a bounded plan-execute-verify-repair loop with multi-provider LLM orchestration (Gemini API with local Ollama fallback) and schema-constrained JSON output.
- Engineered a deterministic safety layer featuring workspace-confined file edits, command risk classification, diff-based approvals, and a rollback journal; validated reliability with 40+ automated unit and integration tests.
- GitHub Repository

MultiThreaded File Processing Engine | C++17, STL, Multithreading, CMake, Linux — Sep 2026

- Engineered a modular C++17 concurrent file-processing engine using a custom thread pool and producer-consumer architecture for high-throughput, low-level I/O execution.
- Implemented a lock-optimized task queue using mutexes, condition variables, `std::future`, and graceful worker shutdown; benchmarked against single-threaded sequential execution.
- GitHub Repository

Agent Ops – Agentic Workflow Builder | Next.js, React Flow, Node.js, WebSockets, Groq API — Jul 2026

- Architected a full-stack agentic AI platform utilizing a ReAct-style loop for autonomous task decomposition and deterministic tool execution.
- Built real-time state streaming and DAG visualization with React Flow and WebSockets; integrated custom tool-calling and Groq API with exponential rate-limit backoff and token-budget enforcement.
- Live Demo

Technical Skills

Languages: C++, Python, JavaScript (ES6+), TypeScript, SQL, HTML/CSS
Core: Data Structures & Algorithms, OOP, Operating Systems, Multithreading, Computer Networks
Frameworks: Node.js, Express.js, FastAPI, React.js, Next.js, Pandas, NumPy, Scikit-learn
Backend & Architecture: RESTful APIs, WebSockets, Microservices, JWT/OAuth2, Redis, Rate Limiting, CI/CD (GitHub Actions)
GenAI / Agentic AI: LLM Orchestration (Gemini, Groq), Agentic Workflows, Prompt Engineering, Speech-to-Text (Whisper)
Databases: MongoDB, PostgreSQL, MySQL, Supabase
Tools: Git, GitHub, Docker, Postman, Linux (Fedora/WSL), CMake, Vercel

Achievements

- **Publication:** Temporal Fusion Transformers for Multi-Horizon Probabilistic Customer Lifetime Value Forecasting (Zenodo DOI: 10.5281/zenodo.19260001)
- **Competitive Programming:** 3-Star Coder on GeeksforGeeks (Rating: 1700+)

Languages

English: Professional Proficiency — Japanese: Intermediate — Hindi: Native